A DUALIZING COMPLEX NORMALISED RELATIVE TO A COVERING IS UNIQUE UP TO UNIQUE ISOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AU7 (sources/stacks-duality/pinned-original-source.tex line 5317).

Allowed standard prerequisites (editorial).

- The sheaf of Homs of two dualizing complexes is invertible. Same chapter, Lemma labelled lemma-dualizing-unique-schemes, cited at proof lines 5347-5350. Makes L invertible.
- The situation and the cocycle gluing condition for a normalised dualizing complex. Same chapter, Situation labelled situation-dualizing, invoked in the statement. Supplies the local trivialisations and the transition maps phi_{ij}.

Source reference. Original reference 0AU4: sources/stacks-duality/pinned-original-source.tex, source lines 5259–5259. Same-chapter pointer into the author environment situation at source line 5259; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATP: sources/stacks-duality/pinned-original-source.tex, source lines 547–547. Same-chapter pointer into the author environment lemma at source line 547; invoked at proof line 5328 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0AU4 let X be a scheme of finite type over S and let $\mathcal{U}$ be a finite open covering of X by schemes separated over S. If there exists a dualizing complex normalized relative to $\omega_S^\bullet$ and $\mathcal{U}$, then it is unique up to unique isomorphism.*

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Proof. If (K, α_i) and (K', α_i') are two, then we consider $L = R\mathcal{H}om_{\mathcal{O}_X}(K, K')$. By Lemma 0ATP and its proof, this is an invertible object of $D(\mathcal{O}_X)$. Using α_i and α_i' we obtain an isomorphism

$$\alpha_i^t \otimes \alpha_i' : L|_{U_i} \longrightarrow R\mathcal{H}om_{\mathcal{O}_X}(\omega_i^\bullet, \omega_i^\bullet) = \mathcal{O}_{U_i}[0]$$

This already implies that $L = H^0(L)[0]$ in $D(\mathcal{O}_X)$. Moreover, $H^0(L)$ is an invertible sheaf with given trivializations on the opens U_i of X. Finally, the condition that $\alpha_j|_{U_i \cap U_j} \circ \alpha_i^{-1}|_{U_i \cap U_j}$ and $\alpha_j'|_{U_i \cap U_j} \circ (\alpha_i')^{-1}|_{U_i \cap U_j}$ both give φ_{ij} implies that the transition maps are 1 and we get an isomorphism $H^0(L) = \mathcal{O}_X$. $\square$